# Arrival-delay prediction with regression
# Micro-Project 3

https://github.com/ChalesManyang/ANA500-Micro-Projects

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# Problem Statement

How accurately can linear regression and support vector regression predict arrival delay after a flight departs?

The continuous outcome is Arrival Delay in Minutes. Predictors available at departure could support better passenger updates.

This project evaluates a retrospective prototype using the supplied airline survey records.

# Hypothesis Formulation

Hypothesis: linear regression and support vector regression will improve RMSE compared with predicting the training-set mean.

Departure delay should explain most of the outcome variation. We also test whether the regression models improve upon simply carrying departure delay forward.

Validation RMSE determines the preferred model before the final test evaluation.

# Acquire

Source: user-supplied airline.csv, loaded with pandas.read_csv.

129,880 passenger records and 25 original columns describe demographics, travel class, service ratings, delays and satisfaction.

The upload has no dates, flight identifiers, routes or sampling documentation. Collection details and the independence of passenger records remain uncertain.

# Prepare

Remove the exported row counter and trim categorical text. No duplicate records or repeated passenger IDs occur.

Exclude 393 missing arrival outcomes, leaving 129,487 records. Seed 42 gives 90,640 train, 19,423 validation and 19,424 test rows.

Predictors: departure delay, distance, age, gender, customer type, travel type and class. Exclude IDs, satisfaction and post-flight service ratings.

Fit numerical scaling and one-hot encoding on training rows only. Keep genuine extreme delays for evaluation.

# Analyze data

Fit ordinary least squares and a scalable linear support vector regressor (LinearSVR). Both use the same training rows and encoded features.

Tune six SVR settings: C = 0.1, 1, 10 and epsilon = 0.01, 0.1 in standardized target units. Use squared epsilon-insensitive loss and a primal solver.

Validation selects C = 0.1 and epsilon = 0.01. Linear regression has slightly lower validation RMSE: 9.8997 versus 9.9023 minutes.

Freeze both models. Clip predictions at zero and evaluate the reserved test set once.

# Report

Test set: 19,424 records. MAE and RMSE use minutes.

Linear regression and SVR have nearly identical RMSE. Both outperform the mean baseline. Carrying departure delay forward has the lowest MAE.

| **Model** | **MAE** | **RMSE** | **$R^2$** |
|---|---|---|---|
| Training mean | 20.76 | 39.16 | -0.000 |
| Departure equals arrival | 5.04 | 10.08 | 0.934 |
| Linear regression | 5.35 | 10.03 | 0.934 |
| Linear SVR | 5.46 | 10.03 | 0.934 |

# Act

Choose linear regression when minimizing squared prediction error is the priority. Its practical advantage over the simple departure-delay baseline is small.

If average absolute error matters most, the departure-delay baseline deserves consideration. Test both options in a flight-level pilot before deployment.

Collect dated flights with route and weather information. Evaluate later flights separately and monitor errors for major delays.

Passenger rows may share a flight, so random splits can overstate generalization. These models predict associations and do not establish causes.